



ACCA

AFM

Advanced Financial Management September 2024

Mock B – Answers



To gain maximum benefit, do not refer to these answers until you have completed the mock B questions and submitted them for marking.

Some of these answers are longer than the examiner would have expected from students in the time available. See the marking schemes to assess how many points were needed to achieve a pass.

IMPORTANT NOTE TO MARKERS (and students):

The AFM exam now contains PROFESSIONAL SKILLS MARKS (10 marks in each 50-mark section A question and 5 marks in each 25-mark section B question).

These are divided into 4 categories:

- communication (which will only be examined in Section A questions)
- analysis and evaluation
- scepticism
- commercial acumen

(Note that not every skill will be tested in every question - see requirements to find out which skills are relevant.)

The communication marks should be easy to award - simply give a maximum of 4 in the Section A question as follows:

1 report format (i.e. intro / subheadings / conclusion)

1 good use of answer tools (i.e. calcs in spreadsheet, writing in word processor)

1 style and clarity of writing

1 following requirement instructions properly (i.e. include everything requirement asks for)

For the other categories, it is a matter of judging how well the student has shown evidence of the skill throughout the answer (for example if asked to talk about the limitations of using a particular model, there is a good chance the student will be able to demonstrate scepticism). At the end of each question, under the marking scheme there is a list of the sort of specific things to look out for.

When adding up the final mark for each question, please show clearly the mark awarded for each professional skill. So for example in the section A question you might award 3/4 for communication, 2/2 for analysis / evaluation, 0/2 for scepticism, and 1/2 for commercial acumen making 6/10 in total. Please display this breakdown clearly.

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SECTION A

1 MARVIN CO

(a) REPORT

To: The Directors, Marvin Co
From: An Advisor
Date: Today
Subject: The proposed Lanavia investment

Introduction

I have evaluated the proposed Lanavia investment by calculating its NPV without considering the follow-on option (see Appendix 1). I have then calculated its 'strategic NPV' incorporating the value of the follow-on option (see Appendix 2).

I have explained the approach taken and my assumptions, before recommending whether the project should be undertaken.

(i) Investment appraisal (ignoring the follow-on option)

As shown in Appendix 1, the NPV of the project ignoring the follow-on option is \$71,000 negative. This negative NPV suggests that the project in isolation is not acceptable.

(ii) Investment appraisal (incorporating the follow-on option)

As shown in Appendix 2, the value of the follow-on option is \$700,000.

Therefore, the overall value of the project ('strategic NPV') incorporating the value of the follow-on option is $\$700,000 - \$71,000$ (from part (i) – Appendix 1) = \$629,000. **(1 mark)**

This positive strategic NPV suggests that the project is in fact acceptable, because the flexibility offered by the option to follow-on is an attractive addition to the basic NPV calculated in part (i) – Appendix 1. **(1 mark)**

(iii) Approach taken **(1 mark per relevant point)**

The approach taken was first to calculate the net present value (NPV) of the initial project based on the given estimates of costs and revenues (without incorporating the follow-on project). The cash flows were discounted at a cost of capital that reflected the risk of the new project.

This was followed by a revised estimate of the 'strategic NPV' value of the project, incorporating the value of the follow-on option.

This was based on viewing the second project as a real option to follow-on (call option) and using the Black–Scholes Option Pricing (BSOP) model to give an estimate of its value.

Assumptions made**(1 mark per relevant point)****Initial project appraisal assumptions**

Unless indicated otherwise, it was assumed that all cash flows occur at the end of the year.

It was assumed that the market research costs of \$500,000 are past costs and therefore not part of the calculation of the value of the project, which is based on future cash flows. **(1 mark)**

It was assumed that all the given estimates such as sales (units and prices), costs, initial investment costs, working capital, and costs of capital and inflation figures are accurate. There is considerable uncertainty surrounding the accuracy of these and a small change in them could change the forecast value of the project quite considerably. A number of projections using sensitivity and scenario analysis would aid in the decision making process.

No details are provided on why the initial project ends after five years. This is an assumption which is made, but the project may last beyond five years and therefore may yield a positive net present value. Additionally, even if the project ceases after five years, it seems unlikely that the 'state-of-the-art' machines and equipment will have such a small realisable value. A larger residual value of these assets could have a considerable bearing on the outcome of the project.

It was assumed that working capital only increases at the rate of inflation, even though the sales levels increase at a much faster rate. More working capital investment might be needed to support the growing level of sales.

The given cost of capital of 10% is apparently appropriate to the project's level of risk but without further explanation on how it has been calculated it is impossible to assess whether this is accurate. The underpinning basis for how the cost of capital was determined may need further investigation.

Although the scenario states that investment will be made and that the project can start immediately, in reality this may not be possible and Marvin Co may need to factor in possible delays.

It was assumed that future exchange rates will reflect the differential in inflation rates between the respective countries (purchasing power parity theory). However, it is unlikely that the exchange rates will move fully in line with the inflation rate differentials.

Option pricing assumptions**(1 mark per relevant point)**

The option for Marvin Co is the opportunity to undertake a further 'follow-on' project in four years' time. This is a call option (a follow-on option in real options theory). Since the option can be exercised after four years, it was considered to be a European type option and the BSOP model was applied.

The BSOP model makes several assumptions such as perfect markets, constant interest rates and normal distribution of asset prices.

It also assumes that volatility can be assessed and stays constant throughout the life of the project, and that the underlying asset can be traded. Neither of these assumptions would necessarily apply to real options.

Conclusion and recommendation

Based on the cost and revenue estimates provided, the net present value before considering the follow-on option is negative at \$71,000 (Appendix 1). However, after taking into account the value of the call option, the strategic net present value is positive at \$629,000 (Appendix 2).

A positive strategic NPV would normally indicate that a project should be undertaken, since it should increase the wealth of the company's shareholders.

However, Marvin Co will not necessarily obtain the current value of the option. The calculated option value represents the best estimate at the moment of what the follow-on project will be worth. Unfortunately though, because the assumptions of the BSOP model don't all apply in the context of real options, the directors need to treat the value obtained as indicative rather than definitive.

The key point is that the initial project is only worthwhile if the follow-on project can also be undertaken. Therefore, before making a final decision some further analysis of the Lanavian market should be undertaken, to reassure the directors that the market conditions in four years' time are highly likely to be compatible with the follow-on project.

Appendix 1 – NPV calculation**Projected cash flows in Lanavia**

	T0	T1	T2	T3	T4	T5
	LP m	LP m	LP m	LP m	LP m	LP m
Sales revenue (W1)		1.800	3.276	4.234	4.862	4.815
VC (W2)		(0.675)	(1.240)	(1.619)	(1.876)	(1.874)
FC (6% inflation)		(1.500)	(1.590)	(1.685)	(1.787)	(1.894)
(1 mark)						
TAD (1m × 20%)		(0.200)	(0.200)	(0.200)	(0.200)	(0.150)
(1 mark)						
Taxable profit		(0.575)	0.246	0.730	0.999	0.897
Lanavia tax payable (15%) (W3)		0	0	(0.060)	(0.150)	(0.135)
(2 marks)						
Add back TAD		0.200	0.200	0.200	0.200	0.150
		(0.375)	0.446	0.870	1.049	0.912
Factory cash flows	(3.000)					2.800
Machine cash flows	(1.000)					0.050
WC (incremental – up 6% each year)	(0.500)	(0.030)	(0.032)	(0.034)	(0.036)	0.632
(2 marks)						
Net cash flow (LP m)	(4.500)	(0.405)	0.414	0.836	1.013	4.394
Exchange rate (W4)	2.58	2.66	2.73	2.81	2.89	2.98
Net cash flow (\$m)	(1.744)	(0.152)	0.152	0.298	0.351	1.474
Additional \$ contribution (3% inflation)		0.120	0.124	0.127	0.131	0.135
(1 mark)						
Tax (25%)		(0.030)	(0.031)	(0.032)	(0.033)	(0.034)
(1 mark)						
Additional tax on Lanavia profits (W5)		0	0	(0.014)	(0.035)	(0.030)
Net cash flows (\$m)	(1.744)	(0.062)	0.245	0.379	0.414	1.545
DF (10%)	1	0.909	0.826	0.751	0.683	0.621
PV	(1.744)	(0.056)	0.202	0.285	0.283	0.959

Net present value = –\$0.071 million i.e. \$71,000 negative

(1 mark)

Workings:

1 Sales price increases by 5% per year (2 marks)

	T1	T2	T3	T4	T5
Price (LP)	120.0	126.0	132.3	138.9	145.9
Units (000)	15	26	32	35	33
Sales revenue (LP million)	1.800	3.276	4.234	4.862	4.815

2 Variable costs increase by 6% per year (1 mark)

	T1	T2	T3	T4	T5
VC (LP)	45.0	47.7	50.6	53.6	56.8
Units (000)	15	26	32	35	33
Total VC (LP million)	0.675	1.240	1.619	1.876	1.874

3 Losses are carried forward and allowed against future profits for taxation purposes.

4 Using purchasing power parity theory (PPPT) and a spot rate of 2.58 LP to \$1, the exchange rates are forecast to increase by a factor of (1.06/1.03) each year:

(2 marks)

	T1	T2	T3	T4	T5
Exchange rate (LP... to \$1)	2.66	2.73	2.81	2.89	2.98

5 Additional tax on Lanavia profits: 10% will be charged (25% – 15%) on the Lanavia taxable profits: **(2 marks)**

	T1	T2	T3	T4	T5
Taxable profit (LP m)	(0.575)	0.246	0.730	0.999	0.897
Adjustment for loss c/f	0.575	(0.246)	(0.329)	–	–
Net (LP m)	0	0	0.401	0.999	0.897
Exchange rate	2.66	2.73	2.81	2.89	2.98
Net (\$ m)	0	0	0.143	0.346	0.301
Tax (10%)	0	0	0.014	0.035	0.030

Appendix 2 – Follow-on option value calculation

The value of the option to follow-on can be calculated using the Black Scholes formula. The five input factors are:

$$r = 0.06$$

$$s = 0.30$$

$$T = 4 \text{ (years)}$$

$$P_e = \$2 \text{ million} \quad \textbf{(1 mark)}$$

P_a = present value of the cash flows from the project (excluding the initial investment) assuming the option to follow-on is exercised **(1 mark)**

$$= \frac{\$0.3\text{m}}{0.10} \times 0.683 = \$2.049 \text{ million}$$

Note that 10% has been used as a discount rate, since the business risk associated with the follow-on project is very similar to that of the initial project.

This is the value of P_a to go into the Black Scholes formula.



Tutorial note

Look out for the BSOP calculator spreadsheet response option in the exam. You'll need to enter the above five key variables in the spreadsheet to generate the answers shown below.

From the BSOP calculator spreadsheet, we can read off the following values:

$$d_1 = 0.7403$$

$$d_2 = 0.1403$$

$$N(d_1) = 0.7705$$

$$N(d_2) = 0.5558$$

$$\text{Call value} = \$0.70\text{m}$$

(1 mark)

$$\text{Put value} = \$0.23\text{m}$$

The relevant value for the follow-on option (call option) is \$0.70m, or \$700,000.

(b) Behavioural finance terminology (1 mark per relevant point)

Gambler's fallacy – investors have a tendency to believe that the probability of a future outcome changes because of the occurrence of various past outcomes e.g. if the value of a share has risen for seven consecutive days, some investors might sell the shares, believing that the share price is more likely to fall on the next day. This is not necessarily the case.

Anchoring – investors have a tendency to attach or 'anchor' their thoughts to a reference point even though it may have no logical relevance to the decision at hand e.g. investors are often attracted to buy shares whose price has fallen considerably because they compare the current price to the previous high (but now irrelevant) price.

Herd instinct – this is the tendency for individuals to mimic the actions (rational or irrational) of a larger group. There are a couple of reasons why herd behaviour happens. The first is the social pressure of conformity – most people are very sociable and have a natural desire to be accepted by a group. The second is the common rationale that it's unlikely that such a large group could be wrong. This is especially prevalent in situations in which an individual has very little experience.

Confirmation bias – it can be difficult to encounter something or someone without having a preconceived opinion. This first impression can be hard to shake because people also tend to selectively filter and pay more attention to information that supports their opinions, while ignoring or rationalising the rest. This type of selective thinking is often referred to as the confirmation bias.

In investing, the confirmation bias suggests that an investor would be more likely to look for information that supports his or her original idea about an investment rather than seek out information that contradicts it. As a result, this bias can often result in faulty decision making because one-sided information tends to skew an investor's frame of reference, leaving them with an incomplete picture of the situation.

Marking scheme			
			Marks
(a)	(i)	Exchange rates	2
		Sales	2
		Variable costs	1
		Fixed costs	1
		Depreciation	1
		Lanavia tax payable	2
		Ignore sunk costs	1
		Working capital	2
		Additional contribution, net of tax	2
		Additional tax	2
		NPV	1
		Reward technique even if calculation errors exist	—
		Total part (a)(i)	17
	(ii)	Current price variable (Pa) in BSOP formula	1
		Other variables in BSOP formula	1
		Correct use of BSOP spreadsheet – pick out call value	1
		Revised value of project and conclusion	2
		Total part (a)(ii)	5
	(iii)	1 mark per sensible point throughout	—
		Approach taken	Max 2
		NPV assumptions	Max 4
		Option pricing assumptions	Max 3
		Conclusion/recommendation	Max 2
		Total part (a)(iii)	Max 10
(b)		1 mark per sensible point throughout – max 3 per term	Max 8
		Total part (b)	Max 8
		Professional skills marks (see below)	10
		Total	50

Professional skills marks

Communication

(max 4 marks)

General report format and structure (use of headings/sub-headings and an introduction)

Style, language and clarity (appropriate layout and tone of report response, presentation of calculations, appropriate use of the tools)

Effectiveness of communication (answer is relevant, specific rather than general and focused to the requirement)

Analysis and Evaluation

(max 2 marks)

Appropriate use of the data to determine suitable calculations

Appropriate use of the data to support discussion and draw appropriate conclusions

Identification of further analysis, which could be carried out to enable an appropriate recommendation to be made

Demonstration of ability to consider relevant factors applicable to Marvin Co's choices

Scepticism

(max 2 marks)

Effective challenge of information and assumptions supplied and techniques carried out to support any investment decision

Demonstration of the ability to probe into the reasons for issues and problems, including the identification of missing information or additional information, which would alter the decision reached

Commercial acumen

(max 2 marks)

Recognition of external constraints and opportunities as necessary

Maximum 10 marks

SECTION B

2 GRINDALL CO

(a) Proceeds from sale of furniture business

Given forecast FCF value = \$347.6m

Expected sales proceeds (10% premium) = $\$347.6\text{m} \times 1.10 = \382.4m **(1 mark)**

Impact on statement of financial position (\$m)

Profit on sale = $\$382.4\text{m} - \$230\text{m} = \$152.4\text{m}$

Two journal entries are needed to record the sale of the furniture business and the investment of the sale proceeds:

1 Dr Current assets (Cash) 382.4

Cr Non-current assets 230

Cr Reserves/Retained earnings 152.4 (balancing figure)

This records as cash the amount received of \$382.4m that represents a 10% premium on the calculated value, and also a book profit of \$152.4m.

2 Cr Current assets (Cash) 382.4

Dr 5% bonds 200

Dr Current assets (Cash) 50

Dr Non-current assets 132.4

So the cash sale proceeds are used to pay off the bonds, leave \$50m in cash and invest the rest in non-current assets.

\$m	Original	1st journal	2nd journal	Final
Non-current assets	692.4	(230.0)	132.4	594.8
				(1 mark)
Current assets	329.1	382.4	(332.4)	379.1
				(1 mark)
Total assets	1,021.5			973.9
Equity and liabilities				
Share capital	400.0			400.0
Retained earnings	210.9	152.4		363.3
				(1 mark)
Non-current liabilities				
5% bonds	200.0		(200.0)	0
Bank loans	130.0			130.0
Current liabilities	80.6			80.6
Total equity and liabilities	1,021.5			973.9
				(1 mark)

Impact on EPS (\$m)

	Current	Revised	
Predicted post-tax profits in Year 1	141.2	141.2	
Less: profits from furniture business		(35.3)	(1 mark)
Add: interest saved, net of tax (\$200m × 5% × (1 – 0.25))		7.5	(1 mark)
Add: return on additional non-current assets (\$132.4m × 15% × (1 – 0.25))		14.9	(1 mark)
Add: return on additional current assets (\$50m × 3% × (1 – 0.25))		1.1	(1 mark)
Adjusted profits	141.2	129.4	
Number of shares	400m	400m	
Adjusted EPS	\$0.353	\$0.324	(1 mark)

Impact on WACC**Equity beta**

$$V_e = \$1.60 \times 1.05 \times 400\text{m} = \$672\text{m}$$

$$V_d = \$130\text{m}$$

$$\beta_e = 0.8 ((672 + 130 (1 - 0.25)) / 672) = 0.916 \quad \textbf{(1 mark)}$$

Revised cost of equity

$$k_e = 3\% + (8\% \times 0.916) = 10.3\% \quad \textbf{(1 mark)}$$

Revised WACC

$$\text{WACC} = 10.3 (672 / (672 + 130)) + 7 (1 - 0.25) (130 / (672 + 130)) = 9.5\% \quad \textbf{(1 mark)}$$

(b) **(1 mark per relevant point)**

In the recessionary times that Grindall Co is facing, it is pleasing to see that the proposal will significantly reduce the company's gearing level, so that there will be less pressure over the next few years to pay interest.

Also, the fact that the furniture business is more likely to be affected by the recession than the food retailing business means that disposing of the furniture business unit makes strategic sense in the short term.

The WACC is forecast to reduce from 10% to 9.5% if the disposal proceeds. We normally expect that a reduction in gearing will lead to an increase in WACC (according to the theories of Modigliani and Miller (M&M)) but in this case the business risk of Grindall Co is changing along with the financial risk, so it is not valid to consider the M&M theory.

The reduction in WACC indicates that the overall risk facing Grindall Co after the disposal of the furniture business unit is perceived to be lower. A lower cost of capital means that future projects will have higher NPVs and generate more wealth for shareholders.

On the other hand, disposing of the furniture business means that Grindall Co will be less well diversified. As a listed company though, this will probably not be too much of a problem for the shareholders.

Listed companies are often majority owned by large shareholders (such as pension funds) who have diversified portfolios of investments. Therefore the shareholders will already have diversified away any unsystematic risk, so they will not benefit further from diversification within the company.

The disposal will reduce earnings per share (EPS), which will send a negative signal to shareholders.

However, the reduction in EPS is only about 8% (from \$0.353 to \$0.324), because the loss of income caused by selling off the furniture business is almost all compensated for by the reduction in interest paid and the returns on new investments. This should help to keep the shareholders happy.

On balance, despite the reduced diversification and lower EPS after the disposal, it looks like selling off the furniture business is a good decision since it will lower gearing, reduce the pressure on cash in the short term and lower the WACC.

Marking scheme	
	Marks
(a) Expected sale proceeds (given value + 10%)	1
Revised SOFP:	
Correct revised NCA figure	1
Correct revised CA figure	1
Correct revised retained earnings figure	1
1 mark if revised SOFP balances!	1
Revised EPS:	
Original year 1 expected earnings	1
Extra return on new NCA investments	1
Extra return on new CA investments	1
Interest saved	1
EPS (revised earnings/number of shares)	1
Revised WACC:	
Equity beta for food retail	1
Cost of equity for food retail	1
Revised WACC	1
Reward technique even if calculation errors exist	—
Total part (a)	13
(b) Arguments for the sale	4
Arguments against (1 mark per sensible point throughout)	4
Total part (b)	Max 7
Professional skills marks (see below)	5
Total	25

Professional skills marks

Analysis and Evaluation

(max 2 marks)

Appropriate use of the data to determine suitable calculations

Appropriate use of the data to support discussion and draw appropriate conclusions

Appraisal of information objectively to make a recommendation

Scepticism**(max 2 marks)**

Effective challenge of information, evidence and assumptions supplied and, techniques carried out to support key facts and/or decisions

Demonstration of ability to consider all relevant factors applicable to a given course of action

Commercial acumen**(max 2 marks)**

Recommendations are practical and plausible in the context of Grindall Co's situation

Effective use of examples and/or calculations from the scenario information and other practical considerations related to the context to illustrate points being made

Maximum 5 marks**3 RACHIDA****(a)****Tutorial note**

For learning purposes, this answer presents full and detailed calculations, showing % costs and monetary costs for each hedging method. In an exam, you would not need to present all of this detail to get full marks.

(i) Futures hedge

1 Set up the hedge:

(a) Buy/sell? Loan: Sell € futures.

(b) Contracts?: $\text{€}3.25\text{m} / \text{€}0.5\text{m} \times 6 \text{ months} / 3 \text{ months} = 13 \text{ contracts}$
(and use June contracts to cover borrowing starting on 1 June).

(c) State the hedge: Sell 13 € June contracts at an opening price of 96.29. **(2 marks)**

2 The play off:

	Now	Rate agreed	P&L	Expiry of contract
	01/12	01/06	01/06	30/06
Company rates	4.50%	5.25%		
Cash market	3.50% (or 96.50)	4.25% (or 95.75)	(0.75%)	
Futures prices	96.29 Sell	95.72 Buy	0.57%	
Basis at 01/06	$0.21\% \times 1/7 = 0.03$			0.00%

(1 mark)

3 Profit on futures:

$57 \text{ ticks} \times 13 \text{ contracts} \times \text{€}12.50 \text{ tick value} = \text{€}9,263$

(1 mark)

4	The cash flow:	
	Actual interest paid =	
	€3,250,000 × (3.5% + 1% + 0.75%) × 6/12	(85,313)
	Profit on futures	9,263
	Net payment	(76,050)
		(1 mark)

- 5 The effective interest 'lock-in' rate is:
- 100 – (opening futures price + unexpired basis)
- = 100 – (96.29 + 0.03) = 3.68%,
- which means that the rate applicable to Rachida (being base rate + 1%) is 4.68%.
- (N.B.** 4.68% × €3.25m × 6/12 = €76,050 as shown above. This reconciliation of monetary cost and % cost will always hold as long as the number of contracts has not been rounded.)
- The futures contract locks into the effective interest of 4.68%. However, this 4.68% rate is not certain, since the futures price in six months' time might not be three basis points different from the cash market rate, because there might not be a linear decline in basis, i.e. **basis risk** might exist. If basis is different from three basis points, the effective cost of borrowing will also be different. **(1 mark)**

(ii) Options on futures

- 1 Set up the hedge:
- (a) Calls/puts? Loan: Sell € futures: Buy € puts (June expiry date, as above)
- (b) Choose an exercise price: **(1 mark)**
- | Exercise price | Loan interest | Cost of options | Lowest cost |
|----------------|---------------|-----------------|--------------|
| 96.25 | 3.75 | 0.185 | 3.935 |
| 96.50 | 3.50 | 0.280 | 3.780 |
| 96.75 | 3.25 | 0.405 | 3.655 |
- (c) Number of contracts: 13 contracts (already calculated in futures part of the question).
- (d) State the hedge: Buy 13 € June put contracts at an exercise price of 96.75. **(1 mark)**
- 2 Cost of the options:
- 0.405 × 100 = 40.5 ticks
- 40.5 × 12.50 × 13 = (€6,581) **(1 mark)**

3	The cash flows:		
	Actual interest paid (as before)	(€85,313)	
	Cost of options	(€6,581)	
	Profit on futures:		
	Sell 96.75		
	Buy 95.72		
	1.03		
	103 ticks × 12.50 × 13	€16,738	(1 mark)
	Net payment	(€75,156)	(1 mark)

This represents an annual % cost of 4.62% ($75,156 / 3.25m \times 12/6$)

However, the futures price at the time when the option would be exercised/sold is not certain.

If interest rates were to fall rather than increase the option could be allowed to lapse (or sold for time value if any) and advantage taken of the lower cash market borrowing rates. (1 mark)

- (b) A payment of B\$2.15 million is due in three months' time, on 1 March.

Hedging with a forward contract (1 mark)

Euro cost of buying B\$2.15 million with a forward contract =

$$\frac{\$2,150,000}{1.4632} = (\text{€}1,469,382)$$

This would be a certain amount, irrespective of what the actual exchange rate is on 1 March.

Hedging with currency options

- 1 Set up the hedge

- (a) Calls or puts:

Buying B\$ → Selling €s (the 'currency of the contract') → € puts

- (b) Choosing the exercise price:

Find the highest net receipts in B\$ per €.

Sell a €	Cost	Receipt in B\$
B\$/€	B\$/€	B\$/€
B\$1.4500	B\$(0.0156)	B\$1.4344
B\$1.4600	B\$(0.0199)	B\$1.4401
B\$1.4700	B\$ (0.0251)	B\$1.4449

- (c) Contracts?: $B\$2,150,000 \div B\$1.4700 \div 31,250 = 46.8$ – say 47 contracts (and use March contracts to cover the 1 March transaction).

- (d) State the hedge: Buy 47 € March put contracts at an exercise price of B\$1.4700. (2 marks)

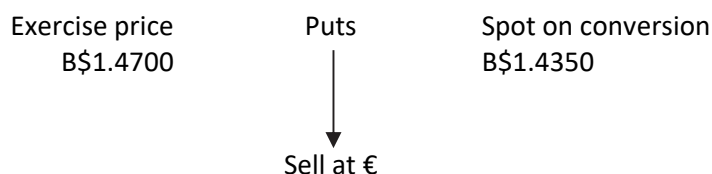
2 Cost of the options – Payable upfront

Option premium = B\$0.0251 × 47 contracts × 31,250 = B\$36,866

Convert at spot – B\$36,866/1.4692 = (€25,093).

(1 mark)

Result of the hedge – should we exercise the option?



The choice is to sell a € under the option at B\$1.4700 or at B\$1.4350 in the spot market. We will sell the € at the highest value hence we will exercise the option.

Cash flows on exercise:

	B\$	€	
Payments in B\$s	(2,150,000)		
Buy B\$s (@B\$1.4700 = €1)			
(€31,250 × 47 × 1.4700)	2,159,063		
Sell €s – € puts		(1,468,750)	i.e. 31,250 × 47
Surplus	9,063		
Sell at spot on conversion			
1.4386		6,300	
Cost of options		(25,093)	
Payments in €s		(1,487,543)	(2 marks)



Tutorial note

In the real exam, you would score most (if not all) of the marks for stating that the surplus (under-hedge or over-hedge) is not material and can therefore be ignored.

The surplus dollars will be sold at the spot rate in three months, because it is not until then that the company knows whether or not the options will be exercised.

(1 mark)

This is much more expensive than hedging in the forward market.

(1 mark)

Marking scheme	
	Marks
(a) Futures	
Sell June futures	1
No. of contracts	1
Hedge, with outcome	3
For full marks basis must be included	
Options on futures:	
Buy 13 June puts	1
Premium cost	1
Overall cost with options	3
Conclusion	1
Comment about allowing the option to lapse if favourable movement	1
	—
Total part (a)	12
(b) Forward contract	1
Buy March puts	1
Correct number of contracts	1
Option premium	1
Outcome at B\$1.4350	2
Discussion and conclusion	2
	—
Total part (b)	8
	—
Professional skills marks (see below)	5
	—
Total	25
	—

Professional skills marks**Analysis and Evaluation****(max 3 marks)**

Appropriate use of the data to determine suitable calculations

Appropriate use of the data to support discussion and draw appropriate conclusions

Appraisal of information objectively to make a hedging recommendation

Commercial acumen**(max 3 marks)**

Effective use of examples and/or practical considerations related to the context to illustrate points being made relating to hedging the interest rate and exchange rate risk

Maximum 5 marks